MonTi with Staged Role Division

A reproduction and development study

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Abstract. We study a staged extension of MonTi for graph-based fraud detection. A frozen upstream generator determines auxiliary-node features and target connections. A separately trained downstream scorer completes the connections to surrounding candidate nodes. Both stages share one structural budget. A requested equal split between the two connection roles is projected to the nearest feasible allocation, preserving every case and its original total node and edge counts.

On GossipCop-S split 0, the complete staged scheme raises mean GAGA target miss rate from **53.6486% to 58.1436%**, a **+4.4950 percentage-point** difference. The three seed gains are positive, with a nominal paired 95% interval of **[0.5715, 8.4185] pp**. Random candidate completion reaches 57.0344%; the smaller trained-minus-random difference remains uncertain. The positive result supports the complete staged scheme relative to its native MonTi baseline in this validation study.

The report first establishes the attack task and reproduction context, then explains the role division, budget constraints, protocol and controlled results. All numerical tables derive from the immutable evidence snapshot **70c48d1a74aa4e5b3e1b1c6ce15c068ba8931980**. The scientific execution source, registration, seed outputs and audits are separately identified for traceability.

1 Original MonTi and the reproduction context

A graph detector classifies nodes using their features and relationships. MonTi adds auxiliary nodes, their features and their connections to make a specified target group harder to detect. In GossipCop-S, nodes are news articles; fraud targets are fake-news articles grouped using shared social engagement metadata. A target group is a list of existing fraud nodes, not an isolated graph or a single new node. Groups can overlap.

MonTi's central idea is to represent targets, surrounding candidates and new auxiliary nodes together with a Transformer. A learned generator jointly plans attributes and connections, while adaptive edge allocation lets auxiliary nodes receive different numbers of edges. Candidate selection bounds the context used for generation. This report retains those upstream capabilities and adds a separate completion role. The original formulation is described in Choi, Kim and Whang, AAAI 2025.

Our earlier formal component reproduction compared learned and random candidate selection on a separate full-test cohort with seeds 0, 1 and 2.

Component comparison	Learned MR %	Random MR %
Paper Table 5	43.7000	39.1300
Our frozen reproduction	43.7280	40.1902

The paired difference was +3.5378 pp, with nominal 95% interval [-1.5415, 8.6171] pp: the frozen conclusion remains **inconclusive**. That result and the later staged study use different cohorts and generator families; their absolute rates are not paired before-and-after measurements.

The paper's main Table 4 reports 41.34 +/- 1.90% for GCN-to-GAGA, a different value from its Table 5. The paper and reproduction comparison records this unresolved difference and the implementation's budget differences.

For the staged experiment, the attack generators are **our three terminal eight-epoch legacy MonTi checkpoints**, not the authors' supplied best MonTi checkpoint. The GCN surrogate and GAGA evaluation weights are fixed official detector checkpoints. Historical scoring semantics and the code's incidence-based budget implementation are retained and documented.

2 The staged improvement

The research question is whether a fixed MonTi planner can support a specialized completion policy under the same structural budget. The design assigns explicit permissions to two stages instead of updating the whole attack generator.

Stage	Decisions	Learning in this experiment
Upstream	Auxiliary features and fixed target connections	Frozen MonTi parameters
Downstream	Auxiliary-to-candidate connections	Train the new scorer only
Combined output	One graph with the original total budget	Evaluate the complete scheme

The upstream first generates a case-specific representation and feature tensor. Its frozen scores determine the target-connection block under the registered quota. The downstream sees the auxiliary features and a summary of these actual target connections before choosing surrounding connections. It cannot change features, target edges, candidate identities or the total budget.

The downstream is a **151,872-parameter scorer**, not a second complete MonTi. It projects auxiliary features, the mean features and count of attached targets, candidate features and cached GCN candidate representations into 64-dimensional hidden vectors. Their interactions score auxiliary-candidate pairs. A fixed selection rule converts scores to an exact number of distinct edges.

Role division provides an operational analogy: a planner commits a partial solution and a specialist completes it. Here the roles are algorithmic permissions, rather than observed identities of real criminals. The experiment tests the complete resulting scheme; its main comparison does not separately identify every component's causal contribution.

Only downstream parameters enter the optimizer. The fixed GCN supplies TRAIN feedback using the mean softplus of the fraud-minus-benign score. Hard graphs are binary; a selected-edge straight-through surrogate supplies approximate gradients during learning. Neither the upstream nor detector weights update.

3 Feasible allocation and matched controls

Equal role budgets are not always geometrically possible. With b auxiliary nodes and t distinct targets, only b*t different auxiliary-to-target pairs exist. Adding duplicate edges cannot legitimately satisfy a larger quota.

Let c be the actual candidate count and e the original total new undirected edge budget. The requested target quota is ceil(e/2). Keeping at least one target edge per auxiliary node gives the feasible interval:

lower = max(b, e - b*c); upper = min(b*t, e).

The target quota is the requested value projected into this interval; the candidate quota is e minus the target quota. An empty interval stops preparation before training. All cases in this run remained feasible and retained their complete denominators.

Across 192 validation seed/group cases, **150 required projection**. Actual target fractions range from **6.3830% to 100%**; the case median is 25.5319%. Only 24 cases have an exact equal split. This is therefore a closest-feasible allocation experiment, rather than an exact-half experiment.

Arm	Target connections	Candidate connections
Native MonTi	Native output	Native output
Quota original	Frozen quota block	Original frozen scores
Quota random	Same frozen quota block	Fixed private random selection
Staged trained	Same frozen quota block	Trained downstream scores

The three quota arms share byte-identical auxiliary features and target blocks, the same candidate lists, quotas and total node/edge budgets. They contain no auxiliary-to-auxiliary edges; native MonTi retains its native structure. Thus native-to-staged changes include allocation and edge-role policy. Individual auxiliary degrees may change even though global structural costs are constant. The matched quota controls help distinguish these effects from learned scoring.

4 Protocol and measurement

The dataset is the official processed GossipCop-S split 0 graph, with news features of dimension 768. We preserve the original group memberships and TRAIN/VALID partitions. The 64 validation groups were selected by a fixed ID-only draw excluding four earlier context lists. They contain 571 target occurrences and 240 unique targets; 233 identities appeared in earlier contexts.

Setting	Registered value
Generator and downstream seeds	3, 4, 5
TRAIN groups	128
VALID groups per seed	64
Downstream training	Four epochs; final checkpoint only
Optimizer	Adam; learning rate 0.001; weight decay 0.0001
Gradient clipping	Norm limit 5
Execution	CPU; four threads
Training feedback	Frozen official GCN
Terminal evaluation	Frozen official GAGA and GCN

All 576 upstream TRAIN/VALID captures were saved before downstream fitting. Final weights and validation topologies were sealed before loading GAGA for evaluation. There was no GAGA checkpoint selection or performance-based early stop. Each seed performed 500 updates; 36 no-choice training occurrences were skipped across the nominal 1,536 opportunities. No cases were dropped from the registered VALID cohort.

MR = 100 * incorrect fraud-target occurrences / all target occurrences.

All target labels are fraud, so an incorrect target prediction is a missed fraud. Auxiliary and surrounding nodes are not added to this denominator. MR weights groups by their number of target occurrences. A group of four targets with one missed target has 25% MR; pre-existing clean misses are also counted.

We compute paired differences across three seeds. The nominal interval is mean difference plus or minus 4.30265 times its sample standard deviation divided by sqrt(3). The 1,713 pooled occurrences are correlated observations, not 1,713 independent replications.

5 Primary validation result

The complete staged scheme improves GAGA MR over native MonTi in every seed.

GAGA arm	Seed 3 MR %	Seed 4 MR %	Seed 5 MR %	Mean MR %
Native MonTi	52.5394	51.4886	56.9177	53.6486
Quota original	56.7426	56.9177	56.9177	56.8593
Quota random	56.7426	57.0928	57.2680	57.0344
Staged trained	58.6690	55.8669	59.8949	**58.1436**

Trained minus control	Mean difference pp	Nominal paired 95% interval pp
Native MonTi	**+4.4950**	**[0.5715, 8.4185]**
Quota original	+1.2843	[-3.9060, 6.4746]
Quota random	+1.1092	[-3.9892, 6.2075]

The native comparison has seed gains of **+6.1296, +4.3783 and +2.9772 pp**. It satisfies the registered screen: at least +1 pp mean gain, all three seed differences positive, a positive lower nominal interval endpoint, and a mean above both quota controls. This is positive evidence for the full staged scheme relative to its native baseline on the registered validation cohort.

The quota controls also improve on native MonTi, showing that allocation and edge-role changes account for part of the observed gain. The trained arm has the highest mean, but its extra advantage over random completion is not stable across seeds: seed 4 is lower by 1.2259 pp. The wide secondary interval does not establish a reliable win over random completion. The complete scheme's positive primary result and this more limited attribution are both retained.

6 Error transitions and training feedback

To distinguish existing weaknesses from attack effects, we separately count clean errors, newly missed targets and recoveries. A new miss changes a clean-correct target to incorrect; a recovery reverses a clean error.

Attacked wrong = clean wrong + new misses - recoveries.

GAGA arm	Clean wrong	Attacked wrong	New misses	Recoveries
Native MonTi	666	919	253	0
Quota original	666	974	308	0
Quota random	666	977	311	0
Staged trained	666	996	330	0

Each row pools the same 1,713 repeated seed/group/node occurrences. The staged scheme has **77 additional wrong occurrences over native MonTi** and 19 over random completion. These are net totals, not counts of independent articles. The unchanged clean denominator prevents a pre-existing miss from being presented as newly caused by the attack.

GCN arm	Seed 3 MR %	Seed 4 MR %	Seed 5 MR %	Mean MR %
Native MonTi	96.8476	98.0736	96.4974	97.1395
Quota original	97.7233	97.8984	97.3730	97.6649
Quota random	97.7233	97.8984	97.3730	97.6649
Staged trained	97.7233	97.8984	97.3730	97.6649

GCN rates are near saturation and equal across the three quota arms. Equal aggregate rates do not imply identical individual predictions. GAGA's separate architecture supplies the more discriminating transfer evaluation here. The result demonstrates additional exposure of a detector under the complete staged intervention; it does not itself provide a defense or establish why each individual connection changes a decision.

7 Verification cost and conclusion

Saved-logit and separately authored saved-tensor audits passed. They recount predictions, transitions, complete denominators, hashes, node identities, projected quotas and owned-edge geometry. The latter verifies stored bindings, rather than independently inspecting every executed DGL graph. Registered TRAIN checks also confirmed finite downstream gradients, parameter updates and unchanged upstream/detector states.

Main execution used 576 upstream generations, 1,500 GCN training forwards, 832 GCN validation forwards and 832 GAGA validation forwards. Main wall time was **3,574.04 seconds**, with **0 GPU hours** and no external API calls. The extra 151,872 parameters and training time are additional computational costs, despite unchanged node/edge budgets. Two startup failures before any real-data model query and their 17.84 guardian seconds remain recorded; all-attempt active guardian time was 3,619.56 seconds. Shared-server timing is descriptive.

Conclusion. Explicit role division between a frozen MonTi planner and a trained candidate-edge completer produced a consistent three-seed gain over native MonTi in this validation study. The +4.4950 pp primary improvement supports the complete staged scheme at fixed total structural budgets. The matched controls show that quota and edge-role changes are material, while the extra learned advantage over random completion remains uncertain.

Evidence	Immutable source at snapshot 70c48d1
Structured result	Summary
Seed and transition counts	Seed metrics
Per-case allocations	Quota records
Exact protocol and inputs	Registration
Audit attribution and byte bindings	Provenance

Scientific execution source: 9e013cb182b5af71a9d0b719b5a939deda3c88e2. Protocol SHA256: 5174bd650628b3300fcf6c091101d176cda304e4136a6801845c421ed451df13. Additional studies and historical evidence are indexed in Supplementary studies.

This is a three-seed development result under an adaptively developed setup, with substantial target-identity reuse and no final TEST evaluation of the staged scheme. A full fair extension against the frozen reproduction baseline within the instructor-approved level remains unconfirmed. Human technical review, instructor approval and actual member contributions are not established by automated evidence; planned duties remain in the contribution record. Codex assisted implementation, experiment supervision, saved-record audits and writing; the AI-use record documents those roles.